

Association Between Urinary Cadmium and Blood Pressure Among U.S. Adults: A Survey-Weighted Analysis of NHANES 2017–2018

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Purpose: Application writing sample / independent health data analysis project

Focus areas: Health data science; environmental epidemiology; biostatistics; survey-weighted regression; R programming

Abstract

Background: Cadmium is a toxic environmental metal with potential cardiovascular effects. Urinary cadmium is commonly used as a biomarker of cumulative cadmium exposure, but its association with blood pressure in the general U.S. adult population remains uncertain.

Methods: This cross-sectional study used data from the 2017–2018 National Health and Nutrition Examination Survey. The analytic sample included adults aged 20 years or older with complete data on urinary cadmium, blood pressure, demographic characteristics, socioeconomic variables, body mass index, smoking variables, serum cotinine, urinary creatinine, and survey design variables. Urinary cadmium was log₂-transformed and modeled as a continuous exposure; therefore, regression coefficients represent the difference in blood pressure associated with each doubling of urinary cadmium concentration. Survey-weighted linear regression models were used to evaluate the association between urinary cadmium and systolic blood pressure. Models were sequentially adjusted for demographic, socioeconomic, anthropometric, smoking-related, and urinary dilution covariates. Sensitivity analyses evaluated diastolic blood pressure, urinary cadmium quartiles, and an unweighted systolic blood pressure model.

Results: The final analytic sample included 1,361 adults. The weighted median age was 50 years (IQR: 35–61), the median urinary cadmium concentration was 0.181 µg/L (IQR: 0.090–0.325), the median systolic blood pressure was 122.7 mmHg (IQR: 113.3–134.7), and the median diastolic blood pressure was 73.3 mmHg (IQR: 66.0–81.0). In the unadjusted survey-weighted model, each doubling of urinary cadmium was associated with a 1.32 mmHg higher systolic blood pressure, although the association did not reach conventional statistical significance (95% CI: -0.09, 2.73; $p = 0.065$). After demographic adjustment, the association was attenuated and changed direction ($\beta = -0.25$; 95% CI: -1.79, 1.29; $p = 0.742$). In the fully adjusted model, each doubling of urinary cadmium was associated with a -0.59 mmHg difference in systolic blood pressure (95% CI: -1.95, 0.77; $p = 0.355$). Sensitivity analyses using diastolic blood pressure, cadmium quartiles, and an unweighted systolic blood pressure model showed broadly consistent null

findings.

Conclusions: In this survey-weighted analysis of NHANES 2017–2018 adults, urinary cadmium was not significantly associated with systolic or diastolic blood pressure after adjustment for demographic, socioeconomic, anthropometric, smoking-related, and urinary creatinine covariates. This project demonstrates a reproducible analytic workflow for environmental epidemiology using publicly available national health survey data.

Keywords: cadmium; blood pressure; NHANES; environmental epidemiology; survey-weighted regression; urinary biomarkers; cardiovascular health

Project Overview

This project was designed as an independent R-based environmental epidemiology analysis using publicly available NHANES 2017–2018 data. The main objective was to evaluate whether urinary cadmium, a biomarker of cumulative cadmium exposure, was associated with blood pressure among U.S. adults.

The project emphasized practical skills relevant to health data science, epidemiology, biostatistics, and research assistant roles, including:

- public health dataset cleaning and merging;
 - complex survey design specification;
 - survey-weighted regression modeling;
 - covariate adjustment;
 - sensitivity analysis;
 - manuscript-style table and figure generation;
 - reproducible R programming;
 - scientific interpretation of epidemiologic findings.
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Data Source and Analytic Sample

The analysis used the 2017–2018 cycle of the National Health and Nutrition Examination Survey. NHANES combines interviews, physical examinations, and laboratory testing using a complex, multistage probability sampling design.

The following NHANES components were merged by participant identifier:

- demographic data;

- blood pressure examination data;
- urinary metal biomarkers;
- body measurement data;
- smoking questionnaire data;
- serum cotinine;
- urinary creatinine;
- survey design variables.

The analytic sample was restricted to adults aged 20 years or older with complete data for the exposure, outcomes, covariates, and survey design variables. The final complete-case analytic sample included **1,361 adults**.

Methods

The primary exposure was urinary cadmium concentration, measured in $\mu\text{g/L}$. Because urinary cadmium was right-skewed, it was log2-transformed for regression modeling. This allowed model coefficients to be interpreted as the difference in blood pressure associated with each doubling of urinary cadmium concentration.

The primary outcome was systolic blood pressure, measured in mmHg and calculated as the mean of up to three available systolic blood pressure readings. Diastolic blood pressure was analyzed as a secondary outcome in sensitivity analysis.

Survey-weighted linear regression models were fitted using NHANES MEC examination weights, strata, and primary sampling units. Three sequential models were used:

- **Model 1:** unadjusted model with log2 urinary cadmium only;
- **Model 2:** adjusted for age, sex, and race/ethnicity;
- **Model 3:** fully adjusted for age, sex, race/ethnicity, family income-to-poverty ratio, education, body mass index, smoking status, log-transformed serum cotinine, and urinary creatinine.

Sensitivity analyses included:

- fully adjusted survey-weighted model using diastolic blood pressure as the outcome;
- urinary cadmium quartile models for systolic blood pressure;
- unweighted fully adjusted linear regression for systolic blood pressure.

Key Results

The final analytic sample included 1,361 adults. The weighted median age was 50 years, the weighted median urinary cadmium concentration was 0.181 µg/L, the weighted median systolic blood pressure was 122.7 mmHg, and the weighted median diastolic blood pressure was 73.3 mmHg.

In the unadjusted survey-weighted model, each doubling of urinary cadmium was associated with a 1.32 mmHg higher systolic blood pressure, but this association was not statistically significant. After demographic adjustment, the estimate was attenuated and changed direction. In the fully adjusted model, urinary cadmium was not significantly associated with systolic blood pressure.

Sensitivity analyses using diastolic blood pressure, urinary cadmium quartiles, and an unweighted systolic blood pressure model also did not show statistically significant associations. Overall, the findings suggested that urinary cadmium was not independently associated with blood pressure after covariate adjustment in this analytic sample.

Table 1. Survey-weighted baseline characteristics of NHANES 2017–2018 adults

Unweighted analytic sample: n = 1,361

Weighted population estimate: N = 63,874,281

Characteristic	Value
Age, years	50 (35, 61)
Sex	
Male	662 (49%)
Female	699 (51%)
Race/ethnicity	
Mexican American	182 (9.3%)
Other Hispanic	116 (6.0%)
Non-Hispanic White	494 (64%)
Non-Hispanic Black	297 (10%)
Non-Hispanic Asian	197 (5.5%)

Other Race	75 (5.1%)
Family income-to-poverty ratio	3.04 (1.52, 5.00)
Education	
<High school	251 (11%)
High school	352 (30%)
Some college	443 (31%)
College+	315 (28%)
Body mass index, kg/m ²	29 (25, 34)
Smoking status	
Never smoker	765 (56%)
Ever smoker	596 (44%)
Serum cotinine	0 (0, 1)
Urinary creatinine	100 (57, 155)
Urinary cadmium, µg/L	0.18 (0.09, 0.33)
Systolic blood pressure, mmHg	123 (113, 135)
Diastolic blood pressure, mmHg	73 (66, 81)

Note: Values are weighted medians (Q1, Q3) for continuous variables and unweighted counts with weighted percentages for categorical variables.

Table 2. Survey-weighted association between urinary cadmium and systolic blood pressure

Model	Exposure	Beta	95% CI	p-value
Model 1: Unadjusted	Log2 urinary cadmium	1.32	-0.09, 2.73	0.065
Model 2: Demographic-adjusted	Log2 urinary cadmium	-0.25	-1.79, 1.29	0.742
Model 3: Fully adjusted	Log2 urinary cadmium	-0.59	-1.95, 0.77	0.355

Note: Beta estimates represent the difference in systolic blood pressure in mmHg associated with each doubling of urinary cadmium concentration.

Table 3. Sensitivity analyses

Sensitivity analysis	Exposure	Beta	95% CI	p-value
DBP outcome, weighted	Log2 urinary cadmium	0.37	-0.35, 1.08	0.273
Cadmium quartiles, weighted	Cadmium quartile Q2 vs Q1	-0.30	-4.19, 3.59	0.871
Cadmium quartiles, weighted	Cadmium quartile Q3 vs Q1	-1.20	-4.85, 2.45	0.484
Cadmium quartiles, weighted	Cadmium quartile Q4 vs Q1	-2.51	-7.48, 2.46	0.282
SBP outcome, unweighted	Log2 urinary cadmium	-0.82	-1.72, 0.07	0.071

Note: For continuous urinary cadmium models, beta estimates represent the difference in blood pressure in mmHg associated with each doubling of urinary cadmium concentration. For quartile models, beta estimates represent differences in systolic blood pressure compared with quartile 1.

Figures

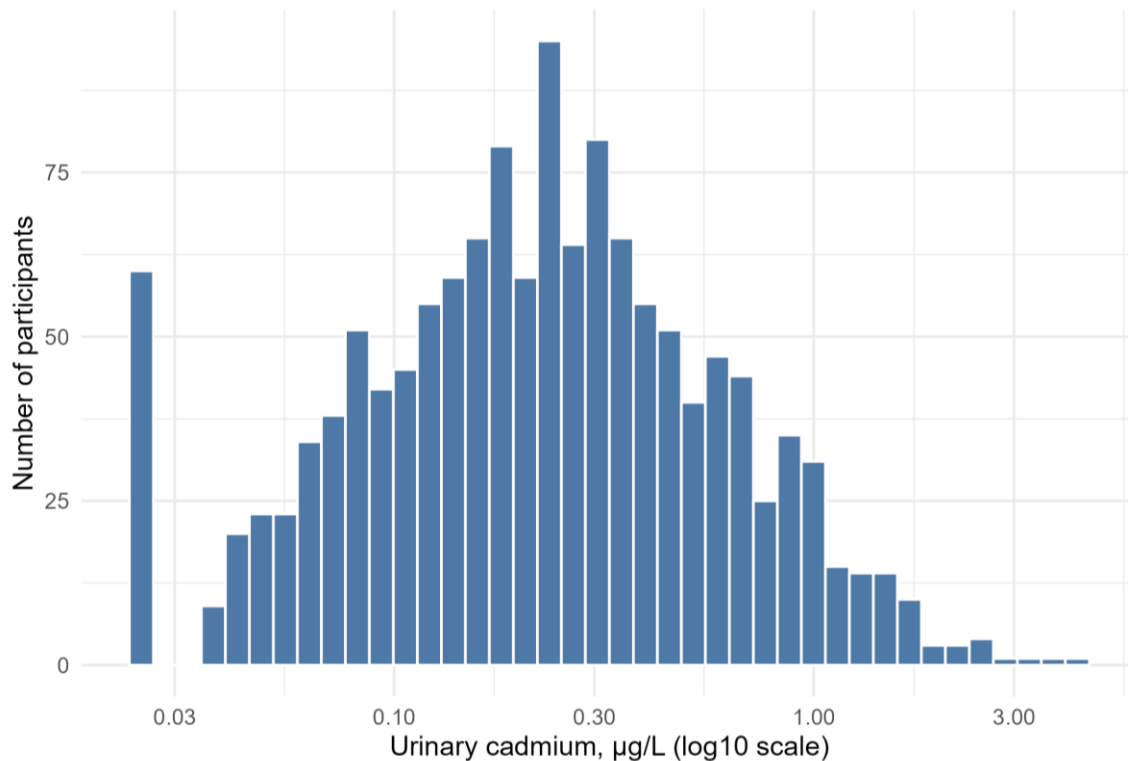


Figure 1. Distribution of urinary cadmium among NHANES 2017–2018 adults. Urinary cadmium concentrations are shown on a log10 scale because the distribution was right-skewed.

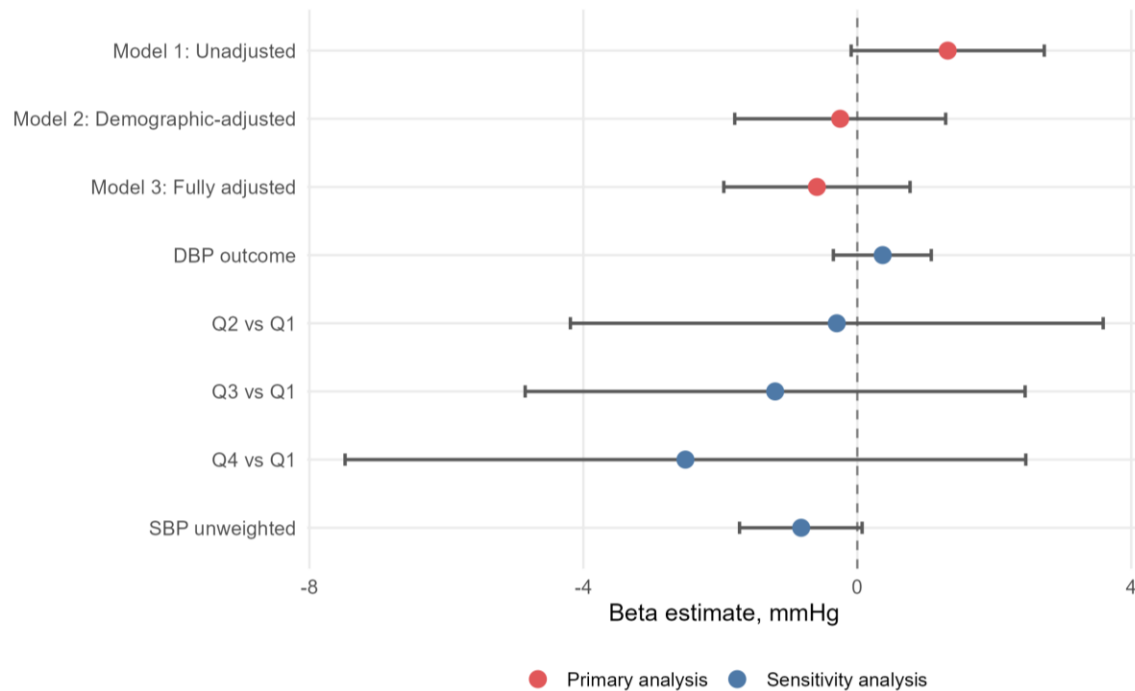


Figure 2. Association between urinary cadmium and blood pressure.

Points represent beta estimates, and horizontal lines represent 95% confidence intervals. Red points show the primary survey-weighted models for systolic blood pressure. Blue points show sensitivity analyses, including diastolic blood pressure, cadmium quartile models, and an unweighted systolic blood pressure model.

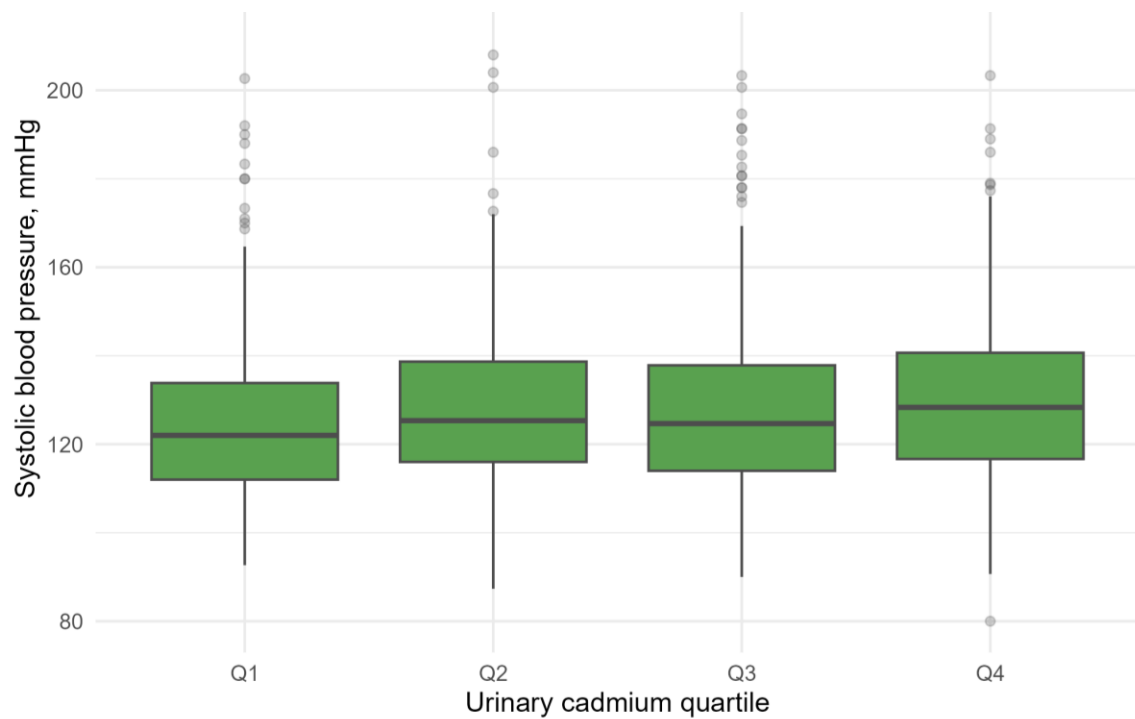


Figure 3. Systolic blood pressure by urinary cadmium quartile.

Boxplots show the median, interquartile range, and outlying values of systolic blood pressure across quartiles of urinary cadmium. This figure is descriptive and does not incorporate survey weights.

Reproducible R Workflow

The R workflow included:

- importing raw NHANES files;
 - cleaning demographic, examination, laboratory, smoking, urinary biomarker, and survey design variables;
 - merging datasets by participant identifier;
 - constructing a complete-case analytic sample;
 - creating transformed exposure variables;
 - specifying the NHANES complex survey design;
 - fitting survey-weighted regression models;
 - conducting sensitivity analyses;
 - generating manuscript-style tables;
 - exporting figures and model results;
 - preparing a writing sample for application purposes.
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Skills Demonstrated

This project demonstrates applied skills in:

- R programming for public health data analysis;
- environmental epidemiology;
- health data cleaning and merging;
- NHANES data structure and survey design;
- survey-weighted linear regression;
- covariate adjustment and confounding assessment;

- sensitivity analysis;
 - data visualization;
 - scientific writing;
 - reproducible analytic workflow development.
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Interpretation and Limitations

The unadjusted model suggested a positive association between urinary cadmium and systolic blood pressure, but this estimate was attenuated after covariate adjustment and was not statistically significant in the fully adjusted model. This pattern suggests that the crude association may have been influenced by demographic, socioeconomic, smoking-related, or urinary dilution-related factors.

The main limitations include the cross-sectional study design, use of one NHANES cycle, spot urine biomarker measurement, complete-case analysis, and potential residual confounding. Future work could pool multiple NHANES cycles, incorporate kidney function biomarkers and antihypertensive medication use, evaluate hypertension as a binary outcome, and examine effect modification by age, sex, smoking status, kidney function, or race/ethnicity.

Code Availability

R scripts, generated tables, figures, and analytic workflow files are available upon request.

Application Project Summary

I conducted a reproducible environmental epidemiology project using NHANES 2017–2018 data to evaluate the association between urinary cadmium and blood pressure among U.S. adults. I cleaned and merged demographic, blood pressure, urinary metal, anthropometric, smoking questionnaire, serum cotinine, and urinary creatinine datasets. I then created a complete analytic dataset, defined the NHANES survey design using sampling weights, strata, and primary sampling units, and implemented survey-weighted regression models in R.

The primary exposure was log2-transformed urinary cadmium, interpreted as the difference in blood pressure associated with each doubling of urinary cadmium concentration. The primary outcome was systolic blood pressure. I fitted sequential

survey-weighted models, including an unadjusted model, a demographic-adjusted model, and a fully adjusted model. The fully adjusted model included age, sex, race/ethnicity, family income-to-poverty ratio, education, body mass index, smoking status, serum cotinine, and urinary creatinine. I also conducted sensitivity analyses using diastolic blood pressure, urinary cadmium quartiles, and an unweighted systolic blood pressure model.

The unadjusted model suggested a positive association between urinary cadmium and systolic blood pressure, but this estimate was attenuated after covariate adjustment and was not statistically significant in the fully adjusted model. Sensitivity analyses were broadly consistent with the primary results and did not provide evidence of a significant association between urinary cadmium and blood pressure after adjustment.

This project generated a cleaned analytic dataset, reproducible model objects, manuscript-style tables, and figures. It strengthened my practical skills in health data analysis, environmental epidemiology, complex survey design, regression modeling, reproducible R programming, and scientific writing using publicly available national health survey data.
